St. Xavier's Collegiate School, Kolkata
First Semester Block Test

Class: 9 Subject: Computer Applications
Date: 14.09.2022

Time: 2 Hrs
Full Marks: 100

Section A
Answer all questions

[20]

Question 1
Choose the correct answer

i) Which of the following is not a Java primitive type
a) byte b) float c) character d) long

ii) What is byte code in Java?
a) Code generated by a Java compiler b) Code generated by a Java Virtual Machine
c) Name of Java source code file d) Block of code written inside a class

iii) What is the final value stored in variable x ?
double x = Math.abs(Math.floor(-8.35));
a) 9.0 b) 7.0 c) 6 d) 7

iv) The number of bytes occupied by char data type is _____byte/s
a) 4 b) 8 c) 2 d) 1

v) Which of the following will give the value **-6.0** :
a) Math.ceil(-6.7) b) Math.abs(-6.7) c) Math.floor(-6.7) d) Math.round(-6.7)

vi) Which datatype cannot be used in a switch case construct?
a) int b) char c) String d) double

vii) Which is not a valid datatype?
a) integer b) short c) long d) double

viii) Evaluate Math.pow(Math.sqrt(25),3);
a) 75.0 b) 75 c) 125.0 d) 15.0

ix) Absence of _____ keyword in switch case construct helps "Fall through".
a) default b) case c) break d) continue

x) double x= 3/10 will give the value
a) 0.3 b) 0.0 c) 3.3 d) 0

xi) The smallest unit in a program is known as _____.
a) data type b) literal c) token d) object

xii) The process of binding the data and method together as one unit is called as:
a) Encapsulation b) Inheritance c) Polymorphism d) Dynamic binding

xiii) Division operator has ____ precedence over multiplication operator.
a) equal b) higher c) lower d) none of these

xiv) A valid combination of operators, constants and variables.
a) expression b) statement c) scope d) condition

xv) What type of conversion is happening in the following statement
int d= 'A';
a) explicit conversion b) implicit conversion c) default d)No conversion

xvi) Which of the following does not represent a character literal ?
a) 'x' b) '9' c) "x" d) '_'

xvii) They represent non-graphic characters.
a) keywords b) literals c) escape sequences d) identifiers

xviii) Which of the following is not a relational operator?
a) = b) >= c)== d)!=

xix) Which of the following is not an OOPs principle?
a) Abstraction b) Polymorphism c) Identifier d) Inheritance

xx) _____ is called an instance of a class
a) Object b) Attributes c) State d) Operators

Question 2

i)Give the output of the following program segments:

a) [4]

```
void check() {
int a=10;
switch(a) {
case 10: System.out.print(a++ +" ");
case 11: System.out.print(++a+" ");
case 12: System.out.print(a++ +" ");
break;
default: System.out.println(a++); }
System.out.println(a); }
```

b) [2]

```
int x=6*7/3+6%5-5+6;
System.out.println(x);
```

c) [4]

```
void one() {
int s=0;
for (int i=1;i<=12;i+=3) {
s=s+i;
System.out.print(i+" "); }
System.out.println(s); }
```

ii) Rewrite the following code using the ternary operator [2]

```
if(price>=100)
```

```
dis=10.0/100*price;
else
dis=15.0/100*price;
```
[2]

iii) Write the Java expression for the following:

$z = x^3 + y^3 - \sqrt{x^2 + y^2}$ [4]

iv) Write the equivalent code of the following, replacing if with switch:-

```
public void month(int m)
{
    int day=0;
    if ( m==2)
        day=28;
    else
    {
     if ( m==4 || m==6 || m==9 || m==11 )
        day=30;
     else
        day=31;
    }
}
```
[2]

v) Give the result of the following expression where a=1,b=2 and c=3

a>b && c*3>=(5+a)

Section B

Answer any 4 questions.

Variable descriptions will be required with every program.

[15]

Question 3

Write a menu-driven program, using switch case, to find the perimeter of a square, perimeter of a rectangle and circumference of a circle as per the user's choice. Display an appropriate message for a wrong choice.

(Perimeter of a square = 4*s, perimeter of a rectangle = 2*(l + b), Circumference of a circle=2 π r)

[15]

Question 4

In a post office, the price for sending letters or parcels is charged as per the slab given below. The price is based on the weight of the item with an additional 5% service tax.

Weight of Parcel / Letter	Rate in Rs
First 10 gms	10
Next 20 gms	30
Next 30 gms	70
More than 60 gms	150

Design a class to accept the weight of the parcel or letter. Calculate and display the payable amount referring to the slab values along with the service tax.

Question 5 [15]

Write a program to enter 20 integers. Calculate and print the sum of the odd positive integers and count and print the number of negative integers.

Question 6 [15]

Design a class to check if a given number is a pronic number or not. [A number is said to be pronic

if the product of two consecutive numbers is equal to the number]

Example: $0 = 0 \times 1$

$2 = 1 \times 2$

$6 = 2 \times 3$

$12 = 3 \times 4$

thus, 0, 2, 6, 12... are pronic numbers.

Question 7 [15]

Write a program to accept x and n and print the sum of the following :-

$$1 + \frac{x^2}{1!} + \frac{x^3}{2!} + \ldots + \frac{x^n}{(n-1)!}$$

where $n! = 1 \times 2 \times 3 \times \ldots \times n$

Question 8 [15]

Write a program to print the following pattern :

1
21
321
4321
54321
4321
321
21
1